Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Belovich, Joanne Ph.D.		
Project Title:	An Accoustically Assisted Bioreactor for Terrestrial and Microgravity Applications		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	j.belovich@csuohio.edu	**Fax:**	FY 216-687-9280
PI Organization Type:	UNIVERSITY	**Phone:**	216-687-2556
Organization Name:	Cleveland State University		
PI Address 1:	2121 Euclid Ave.		
PI Address 2:	Engineering Dean's Office SH 104		
PI Web Page:	http://www.csuohio.edu/ ; http://www.csuohio.edu/		
City:	Cleveland	**State:**	OH
Zip Code:	44115	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-02
Start Date:	03/01/1999	**End Date:**	05/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Richmond, Robert	**Contact Phone:**	256 544-3418
Contact Email:	robert.richmond@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feke, Donald (Case Western Reserve University)		
Grant/Contract No.:	NAG8-1584		
Performance Goal No.:	2.9		
Performance Goal Text:	Determine the potential use of bioreactors to provide biological materials supporting long-duration space travel.		

This proposal addresses the need for advanced concepts in bioprocessing for application in both terrestrial and microgravity environments. The small size and fragile nature of the mammalian cells used to produce pharmaceuticals and tissue for organ replacement results in inherent processing difficulties. Specifically, mass transport to and from the cells may be deficient, especially in bioreactors with high cell concentrations, excessive agitation can damage the cells, and hydrodynamic entrainment may result in the loss of viable cells from bioreactors operating in perfusion mode. Recent experiments in our laboratory have demonstrated a technique for the retention of micron-sized solids (in the same size range of mammalian cells) within the millimeter-sized pores of a highly porous medium. The technique involves the generation of an ultrasonic resonant field within the porous medium. The resulting acoustic forces acting on the particles stabilize them relative to the flow of suspending fluid through the porous medium particles. We propose to investigate the usefulness of such an acoustic technique for continuous bioprocessing. Such a configuration has potential advantages

Task Description:	over conventional bioreactors which include enhanced transport rates at the cell surface (since the acoustic field continues to gently agitate the cells), low pressure drops through the reactor (since pore sizes are relatively large), no need for a separation and recycle of viable cells from the reactor effluent stream (since the acoustic field provides the retaining force), and efficient cell harvesting. The proposed bioreactor can function easily in a microgravity environment. The acoustic bioreactor may be used as means for studying tissue growth under microgravity conditions since the open structure of the porous medium will allow visual observation of growing tissue. The first phase of the project is to measure the efficiency of mammalian cell retention and cell viability as a function of the properties of the cell suspension, the configuration, properties, and operating conditions of the acoustic chamber, in order to identify the optimal operating parameters for the bioreactor. A mathematical model which addresses the hypothesis of the mechanism for cell response to acoustic fields will be developed, and experiments will be conducted to verify the models and hypotheses. The results from the modeling will be used to design an efficient acoustic chamber for use as a bioreactor. The acoustic bioreactor will be constructed and evaluated for effectiveness in growing cells and producing antibodies, and the results compared to traditional reactor designs. The results of this highly innovative research can be expected to benefit the field of tissue engineering research and the pharmaceutical industry. The regeneration of tissue often requires growth of the cells in a three-dimensional matrix that mimics the in vivo structure. The acoustic bioreactor design will provide for this three-dimensional geometry, without the mass transfer limitations that are common to multi-layer tissue constructs. This process may therefore improve the ability to expand bone marrow, to generate blood cells from bone marrow, to regenerate liver tissue, and so on. The large-pore nature of the porous medium can allow visual observation of cell growth to aid in the understanding of the tissue growth process. This bioreactor will also be useful for the production of high-value biopharmaceutical products produced by mammalian cells. By removing the extra immobilization step, and increasing mass transfer rates, and increasing cell density in the react, we can expect that processing costs will decrease. The bioreactor design developed here will have applications in space, where local production facilities for biopharmaceuticals may be needed. The acoustic forces are used to counteract the forces that carry the cells with the fluid motion, without the need for filtration screens or gravity. Therefore, this reactor will operate similarly both with and without gravity. The acoustic bioreactor may be used as a means for studying tissue growth under microgravity conditions, since the open structure of the porous medium will allow visual observation of growing tissue and the acoustic fields will prevent the entrainment of the cells in the perfusion media. Furthermore, the suspension of the cells in low-shear flow conditions with ample mass transport may provide the necessary environment for tissue differentiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)